

Let us consider once more the set S from Theorem 2.2.3, together with any matching M in $G = (V, E)$. As before, we write $\mathcal{C} := \mathcal{C}_{G-S}$. Let us denote by k_S the number of edges in M with at least one end in S , and by $k_{\mathcal{C}}$ the number of edges in M with both ends in $G - S$. Since each $C \in \mathcal{C}$ is odd, at least one of its vertices is not incident with an edge of the second type. Therefore every matching M satisfies

$$k_S \leq |S| \quad \text{and} \quad k_{\mathcal{C}} \leq \frac{1}{2}(|V| - |S| - |\mathcal{C}|). \quad (1)$$

Moreover, G contains a matching M_0 with equality in both cases: first choose $|S|$ edges between S and $\bigcup \mathcal{C}$ according to (i), and then use (ii) to find a suitable set of $\frac{1}{2}(|\mathcal{C}| - 1)$ edges in every component $C \in \mathcal{C}$. This matching M_0 thus has exactly

$$|M_0| = |S| + \frac{1}{2}(|V| - |S| - |\mathcal{C}|) \quad (2)$$

edges.

Now (1) and (2) together imply that every matching M of maximum cardinality satisfies both parts of (1) with equality: by $|M| \geq |M_0|$ and (2), M has at least $|S| + \frac{1}{2}(|V| - |S| - |\mathcal{C}|)$ edges, which implies by (1) that neither of the inequalities in (1) can be strict. But equality in (1), in turn, implies that M has the structure described above: by $k_S = |S|$, every vertex $s \in S$ is the end of an edge $st \in M$ with $t \in G - S$, and by $k_{\mathcal{C}} = \frac{1}{2}(|V| - |S| - |\mathcal{C}|)$ exactly $\frac{1}{2}(|\mathcal{C}| - 1)$ edges of M lie in C , for every $C \in \mathcal{C}$. Finally, since these latter edges miss only one vertex in each C , the ends t of the edges st above lie in different components C for different s .

The seemingly technical Theorem 2.2.3 thus hides a wealth of structural information: it contains the essence of a detailed description of all maximum-cardinality matchings in all graphs. A reference to the full statement of this result, the *Gallai-Edmonds structure theorem*, is given in the notes at the end of this chapter.

2.3 The Erdős-Pósa theorem

Much of the charm of König's and Hall's theorems in Section 2.1 lies in the fact that they guarantee the existence of the desired matching as soon as some obvious obstruction does not occur. In König's theorem, we can find k independent edges in our graph unless we can cover all its edges by fewer than k vertices (in which case it is obviously impossible).

More generally, if G is an arbitrary graph, not necessarily bipartite, and $\mathcal{H}$ is any class of graphs, we might compare the largest number k of graphs from $\mathcal{H}$ (not necessarily distinct) that we can pack disjointly into G with the smallest number s of vertices of G that will cover all

*Erdős-Pósa
property*

its subgraphs in $\mathcal{H}$. If s can be bounded by a function of k , i.e. independently of G , we say that $\mathcal{H}$ has the *Erdős-Pósa property*. (Thus, formally, $\mathcal{H}$ has this property if there exists an $\mathbb{N} \rightarrow \mathbb{N}$ function $k \mapsto f(k)$ such that, for every k and G , either G contains k disjoint subgraphs each isomorphic to a graph in $\mathcal{H}$, or there is a set $U \subseteq V(G)$ of at most $f(k)$ vertices such that $G - U$ has no subgraph in $\mathcal{H}$.)

Our aim in this section is to prove the theorem of Erdős and Pósa that the class of all cycles has this property: we shall find a function f (about $4k \log k$) such that every graph contains either k disjoint cycles or a set of at most $f(k)$ vertices covering all its cycles.

We begin by proving a stronger assertion for cubic graphs. For $k \in \mathbb{N}$, put

$$r_k, s_k := \begin{cases} 4kr_k & \text{if } k \geq 2 \\ 1 & \text{if } k \leq 1 \end{cases} \quad \text{where } r_k := \log k + \log \log k + 4.$$

Lemma 2.3.1. *Let $k \in \mathbb{N}$, and let H be a cubic multigraph. If $|H| \geq s_k$, then H contains k disjoint cycles.*

(1.3.5) *Proof.* We apply induction on k . For $k \leq 1$ the assertion is trivial, so let $k \geq 2$ be given for the induction step. Let C be a shortest cycle in H .

m We first show that $H - C$ contains a subdivision of a cubic multigraph H' with $|H'| \geq |H| - 2|C|$. Let m be the number of edges between C and $H - C$. Since H is cubic and $d(C) = 2$, we have $m \leq |C|$. We now consider bipartitions $\{V_1, V_2\}$ of $V(H)$, beginning with $V_1 := V(C)$ and allowing $V_2 = \emptyset$. If $H[V_2]$ has a vertex of degree at most 1 we move this vertex to V_1 , obtaining a new partition $\{V_1, V_2\}$ crossed by fewer edges. Suppose we can perform a sequence of n such moves, but no more. (Our assumptions imply $n \leq 3$, but we do not formally need this.) Then the resulting partition $\{V_1, V_2\}$ is crossed by at most $m - n$ edges. And $H[V_2]$ has at most $m - n$ vertices of degree less than 3, because each of these is incident with a crossing edge. These vertices have degree exactly 2 in $H[V_2]$, since we could not move them to V_1 . Let H' be the cubic multigraph obtained from $H[V_2]$ by suppressing these vertices. Then

$$|H'| \geq |H| - |C| - n - (m - n) \geq |H| - 2|C|,$$

as desired.

To complete the proof, it suffices to show that $|H'| \geq s_{k-1}$. Since $|C| \leq 2 \log |H|$ by Corollary 1.3.5 (or by $|H| \geq s_k$, if $|C| = g(H) \leq 2$), and $|H| \geq s_k \geq 6$, we have

$$|H'| \geq |H| - 2|C| \geq |H| - 4 \log |H| \geq s_k - 4 \log s_k.$$

(In the last inequality we use that the function $x \mapsto x - 4 \log x$ increases for $x \geq 6$.)

It thus remains to show that $s_k - 4 \log s_k \geq s_{k-1}$. For $k = 2$ this is clear, so we assume that $k \geq 3$. Then $r_k \leq 4 \log k$ (which is obvious for $k \geq 4$, while the case of $k = 3$ has to be calculated), and hence

$$\begin{aligned} s_k - 4 \log s_k &= 4(k-1)r_k + 4 \log k + 4 \log \log k + 16 \\ &\quad - (8 + 4 \log k + 4 \log r_k) \\ &\geq s_{k-1} + 4 \log \log k + 8 - 4 \log(4 \log k) \\ &= s_{k-1}. \end{aligned} \quad \square$$

Theorem 2.3.2. (Erdős & Pósa 1965)

There is a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that, given any $k \in \mathbb{N}$, every graph contains either k disjoint cycles or a set of at most $f(k)$ vertices meeting all its cycles.

Proof. We show the result for $f(k) := \lfloor s_k + k - 1 \rfloor$. Let k be given, and let G be any graph. We may assume that G contains a cycle, and so it has a maximal subgraph H in which every vertex has degree 2 or 3. Let U be its set of degree 3 vertices. U

Let $\mathcal{C}$ be the set of all cycles in G that avoid U and meet H in exactly one vertex. Let $Z \subseteq V(H) \setminus U$ be the set of those vertices. For each $z \in Z$ pick a cycle $C_z \in \mathcal{C}$ that meets H in z , and put $\mathcal{C}' := \{C_z \mid z \in Z\}$. By the maximality of H , the cycles in $\mathcal{C}'$ are disjoint. Z

Let $\mathcal{D}$ be the set of the 2-regular components of H that avoid Z . Then $\mathcal{C}' \cup \mathcal{D}$ is another set of disjoint cycles. If $|\mathcal{C}' \cup \mathcal{D}| \geq k$, we are done. Otherwise we can add to Z one vertex from each cycle in $\mathcal{D}$ to obtain a set X of at most $k - 1$ vertices that meets all the cycles in $\mathcal{C}$ and all the 2-regular components of H . Now consider any cycle of G that avoids X . By the maximality of H it meets H . But it is not a component of H , it does not lie in $\mathcal{C}$, and it does not contain an H -path between distinct vertices outside U (by the maximality of H). So this cycle meets U . X

We have shown that every cycle in G meets $X \cup U$. As $|X| \leq k - 1$, it thus suffices to show that $|U| < s_k$ unless H contains k disjoint cycles. But this follows from Lemma 2.3.1 applied to the multigraph obtained from H by suppressing its vertices of degree 2. □

A very short proof of Theorem 2.3.2 as such, with a recursively defined (and much worse) bound, is indicated in Exercise 11 of Chapter 9.

Our proof of Theorem 2.3.2 can be adapted to give an analogous result for packing cycles edge-disjointly and covering them by edges; this is outlined in Exercise 22 of Chapter 7. A simpler proof of the edge version using Ramsey's theorem is indicated in Exercise 12 of Chapter 9.

We shall also meet the Erdős-Pósa property again in Chapter 12. There, a considerable extension of Theorem 2.3.2 will appear as an unexpected and easy corollary of the theory of graph minors.

2.4 Tree packing and arboricity

In this section we consider packing and covering in terms of edges rather than vertices. How many edge-disjoint spanning trees can we find in a given connected graph? And how few trees, not necessarily edge-disjoint, suffice to cover all its edges? These two questions have two classical theorems answering them. But rather than proving these theorems directly, we shall obtain them both as corollaries of a beautiful recent unification due to Bowler and Carmesin: the *packing-covering* theorem.

To motivate the tree packing problem, assume for a moment that our graph represents a communication network, and that for every choice of two vertices we want to be able to find k edge-disjoint paths between them. Menger's theorem (3.3.6) in the next chapter will tell us that such paths exist as soon as our graph is k -edge-connected, which is clearly also necessary. This is a good theorem, but it does not tell us how to find those paths; in particular, having found them for one pair of endvertices we are not necessarily better placed to find them for another pair. If our graph has k edge-disjoint spanning trees, however, there will always be k canonical such paths, one in each tree. Once we have stored those trees in our computer, we shall always be able to find the k paths quickly, between any given pair of vertices.

When does a graph G have k edge-disjoint spanning trees? If it does, it clearly must be k -edge-connected. The converse, however, is easily seen to be false (try $k = 2$); indeed it is not even clear that any edge-connectivity will imply the existence of k edge-disjoint spanning trees. (But see Corollary 2.4.2 below.)

Here is another necessary condition. If G has k edge-disjoint spanning trees, then with respect to any partition of $V(G)$ into r sets, every spanning tree of G has at least $r - 1$ *cross-edges*, edges whose ends lie in different partition sets. (Why?) Thus if G has k edge-disjoint spanning trees, it has at least $k(r - 1)$ cross-edges. This condition is also sufficient:

Theorem 2.4.1. (Nash-Williams 1961; Tutte 1961)

A multigraph contains k edge-disjoint spanning trees if and only if for every partition P of its vertex set it has at least $k(|P| - 1)$ cross-edges.

Theorem 2.4.1 has a striking corollary: $2k$ -edge-connectedness is enough to ensure the existence of k edge-disjoint spanning trees.

Corollary 2.4.2. *Every $2k$ -edge-connected multigraph G has k edge-disjoint spanning trees.*

Proof. Every class in a vertex partition of G is joined to other partition classes by at least $2k$ edges. Hence, for any partition into r sets, G has at least $\frac{1}{2} \sum_{i=1}^r 2k = kr$ cross-edges. The assertion thus follows from Theorem 2.4.1. $\square$

cross-edges

tree
packing
theorem
[8.6.9]

[6.4.4]